

Can intrapartum Sildenafil Citrate safely avert the risks of contraction-induced hypoxia in labour? iSEARCH – a pragmatic multicentre Phase III randomised controlled trial.

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Chief Investigator &

Study Chair: Professor Sailesh Kumar

Co-investigators: Prof Ben W. Mol
Prof William Tarnow-Mordi
Prof Vicki Flenady
Prof Helen Liley
Prof Nadia Badawi
Prof Susan Walker
Prof Jon Hyett
Prof Lisa Askie
A/Prof Emily Callander

Coordinating Centre: NHMRC Clinical Trials Centre
92-94 Parramatta Road
Camperdown, NSW 2050
Telephone: 61-2-9562-5000
Fax: 61-2-9565-1863
Email: iSEARCH.study@sydney.edu.au

Senior statistician: Dr Rachel O'Connell
NHMRC Clinical Trials Centre, University of Sydney

Project Manager: Dr Alpana Ghadge
NHMRC Clinical Trials Centre, University of Sydney

Abbreviations

AE	Adverse Event
AIHW	Australian Institute of Health and Welfare
ASQ	Ages and Stages Questionnaire
BSIDIII	Bayley Scales of Infant Development-III
CPAP	Continuous Positive Airway Pressure
CPR	Cerebroplacental Ratio
CRF	Case report form
CS	Caesarean Section
CTC	NHMRC Clinical Trials Centre, University of Sydney
CTG	Cardiotocography
DNA	Deoxyribonucleic acid
ECG	Electrocardiography
FDA	Food and Drug Administration
GP	General Practitioner
GSA	Group Specific Appendix
HIE	Hypoxic Ischaemic Encephalopathy
HREC	Human Research Ethics Committee
IDSMC	Independent Data, Safety Monitoring Committee
IVB	Instrumental Vaginal Birth
MBS	Medicare benefits Scheme (Australia)
NDI	National Death Index
NICU	Neonatal Intensive Care Unit
NE	Neonatal Encephalopathy
NO	Nitric Oxide
P2	Pulmonic Closure Sound
PBS	Pharmaceutical Benefits Scheme (Australia)
PIGF	Placental Growth Factor
QOL	Quality of life
SAE	Serious adverse event
SC	Sildenafil Citrate
SSI	Significant Safety Issue
SUSAR	Suspected Unexpected Serious Adverse Reaction
TCPO2	Transcutaneous oximetry
TGA	Therapeutic Goods Administration
USM	Urgent Safety Measure

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SYNOPSIS AND SCHEMA

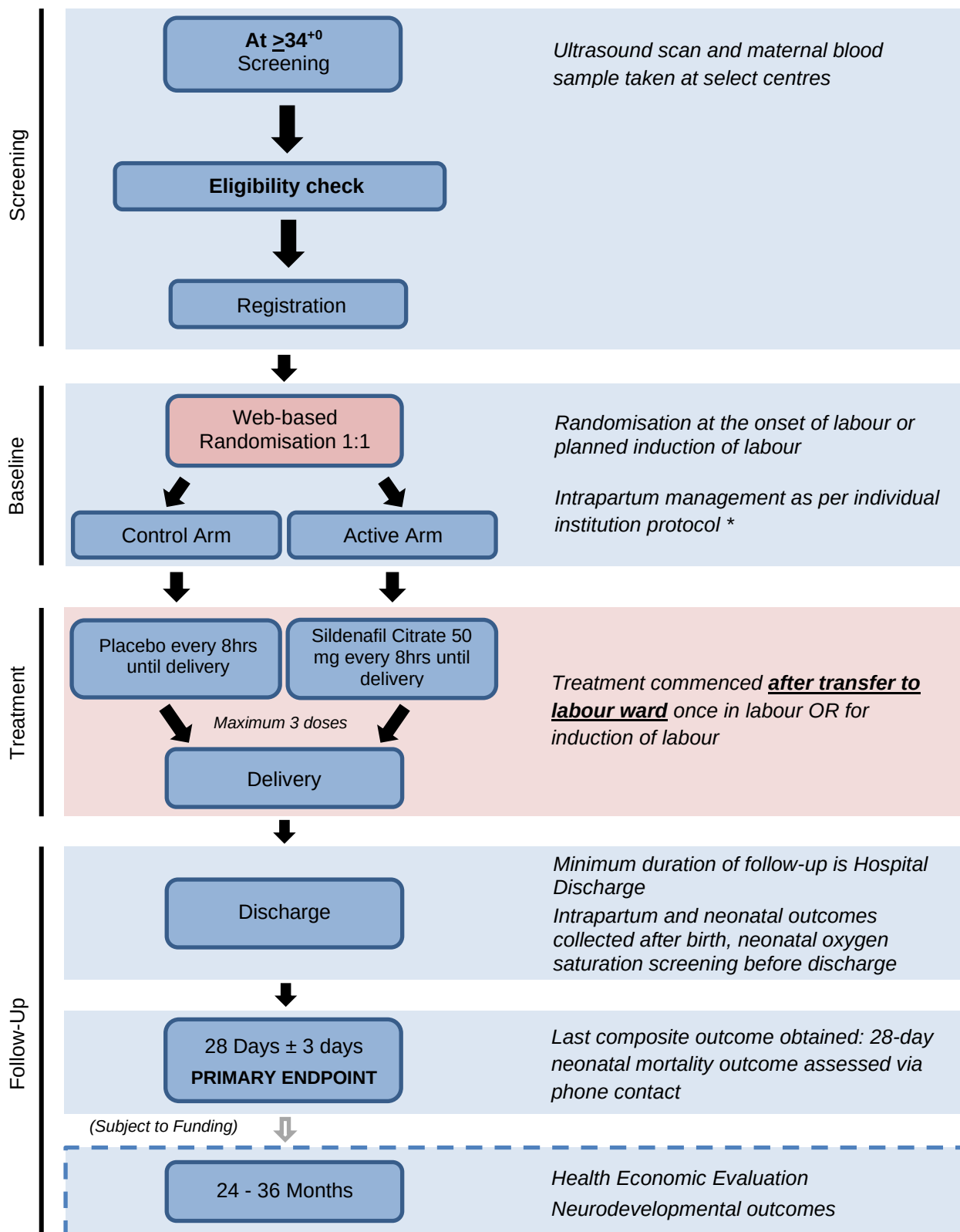
Background	In Australia, almost 1 in 4 caesarean sections are performed for suspected fetal distress. A fetus can become distressed when the placenta is unable to adequately provide it with sufficient oxygen and nutrients during labour. The only therapeutic options when fetal distress develops are emergency operative birth by caesarean section (CS) or instrumental vaginal birth (IVB) – both of which can have serious consequences. Sildenafil Citrate (SC) preferentially dilates pelvic blood vessels and increases utero-placental blood flow. We propose that oral SC improves uterine perfusion in labour thus reducing adverse perinatal outcomes due to fetal distress.
Aim	The iSEARCH Trial will determine if, in women in labour at term <i>receiving</i> oral 50 mg SC vs placebo given in ≤ 3 doses, reduces a primary composite endpoint, determined up to 28 days after birth, of ten perinatal outcomes related to intrapartum hypoxia.
Primary objective	Compared with placebo, SC achieves a 35% reduction from 7% to 4.55% in the relative risk of the composite perinatal outcome, determined 28 days after birth: (i) Intrapartum stillbirth or (ii) 28-day neonatal death, or (iii) Apgar score <4 at 5 minutes, or (iv) cord artery pH <7.0 , or (v) neonatal encephalopathy, or (vi) neonatal seizures, or (vii) neonatal respiratory support for >4 h, or (viii) neonatal unit (Special Care or Intensive Care) admission lasting >48 h, or (ix) persistent pulmonary hypertension of the newborn or (x) meconium aspiration
Secondary objectives	1. Compared with placebo, SC improves the rate of each of the ten items of the primary composite perinatal outcome. 2. Compared with placebo, SC reduces the risk of CS or IVB for fetal distress by 25% (from 20% to 15%). 3. Compared with placebo, SC is cost-effective.
Tertiary and correlative objectives	1. Compared with placebo, SC improves the rate of the following Maternal and Obstetric Outcomes: CS for fetal distress, IVB for fetal distress, CS for other indications, IVB for other indications, spontaneous vaginal birth, length of 1st and 2nd stage labour, post-partum haemorrhage (blood loss >1000ml), post-partum hysterectomy, uterine rupture, Maternal Intensive Care Unit admission, Maternal mortality. 2. Compared with placebo, SC reduces the relative risks of cerebral palsy, cognitive delay and neurosensory disability at 2 years of age.
Design	A randomised, multicentre, two-arm parallel, phase III, double-blind, placebo-controlled trial.
Population	Women with singleton or twin pregnancies, planning vaginal birth at ≥ 37 weeks gestation, screened for eligibility and approached by research midwives from 34 ⁺⁰ weeks for consent to participant in the trial.
Study treatments	The treatment group will receive 50 mg of SC orally every 8 hours (no more than 3 doses) when admitted in spontaneous labour or for induction of labour. The control group will receive matching placebo at the same timepoints (no more than 3 doses).
Assessments	Clinical assessments at baseline, after labour, birth, pre-discharge and Ages and Stages Questionnaire ^{1,2} for neurodevelopmental outcomes at 2 years of age.
Statistical considerations	A sample size of 3,200 is needed to detect a 35% reduction from 7% to 4.55% for an alpha of 0.05 and $>80\%$ power with 10% drop-out in each arm. This sample size would also yield $>90\%$ power to detect a 25% reduction for the secondary outcome CS or IVB for fetal distress.

STUDY SCHEMA

Duration of accrual: 4 years

Stratification: Study Sites

Primary Endpoint: Composite of ten perinatal outcomes related to intrapartum hypoxia, determined 28 days from birth.



* In-hospital maternal and neonatal outcome events occurring from screening to discharge home will be collected from medical records. There are no formal study assessments visits.

1 BACKGROUND

Hypoxia during labour causes fetal distress, a key factor in 23% of emergency caesarean births in Australia.³ Hypoxic injury can lead to intrapartum stillbirth, severe neonatal morbidity, neonatal encephalopathy (NE) and cerebral palsy.^{4 5 6} NE is a major risk factor for cerebral palsy. Eighty percent of infants with moderate or severe NE suffer hypoxic brain injury near the time of birth - in >50% of cases this injury occurs during labour.⁷ Other than emergency operative birth by Caesarean Section (CS) or by Instrumental Vaginal Birth (IVB), we lack therapeutic options to achieve safe outcomes for the infant.

Sildenafil Citrate preferentially dilates pelvic vessels, increasing utero-placental perfusion.⁸ It was originally used in hypertension and ischaemic heart disease and is effective in adult and newborn pulmonary hypertension.^{9 10} In our systematic review of 16 maternal studies, side effects of SC were mild with no long-term adverse outcomes.¹¹

Pilot data from the RidStress Phase 2 RCT

We showed in a world-first Phase 2 RCT (RidStress) of 300 women at term, in a Report of Major Impact in the American Journal of Obstetrics and Gynaecology in January 2020,¹² that up to 3 oral doses of SC vs placebo: (a) halved CS or IVB for fetal distress [RR 0.49, 95% Confidence Interval (CI) 0.33 - 0.73], $p=0.0004$] (b) halved the duration of the 2nd stage of labour (36 vs. 66 min, $p=0.02$) and (c) halved fetal heart rate (FHR) abnormalities [15% vs. 32%; RR 0.48, 95% CI 0.31 - 0.75, $p=0.0009$]. The trial however lacked power to evaluate neonatal outcomes.

In partnership with the Mater Research Institute-University of Queensland, NHMRC Clinical Trials Centre (CTC), Stillbirth Centre for Research Excellence and the Australian Cerebral Palsy Alliance, this MRFF funded, 4-year, Phase 3 RCT will test if SC reduces the relative risk of a primary composite endpoint of ten perinatal outcomes attributable to intrapartum hypoxia by 35%, from 7% to 4.55%, in 3200 women in spontaneous or induced labour at term in 16 Australian hospitals.

iSEARCH is the world's first Phase 3 trial to re-purpose SC - a widely available, well characterised, affordable, off-patent vasodilator drug - by evaluating whether, compared to Placebo, it improves perinatal and maternal outcomes by reducing operative birth for fetal distress. These are major areas of unmet need, identified as critical priorities by The Lancet.^{13 14} iSEARCH will also contribute to a separately funded, international Individual Participant Data Prospective Meta-Analysis (IPD PMA) powered to show a 10% to 20% reduction in intrapartum stillbirth, 28 day neonatal mortality and other severe neonatal morbidity attributable to intrapartum hypoxia. The IPD PMA will include multiple concurrent international trials in high, low- and middle income countries following a shared protocol, funded by relevant national and international bodies and guided by an international Data and Safety Monitoring Committee with an agreed Charter.¹²

The STRIDER Trials: The international STRIDER (Sildenafil Therapy In Dismal prognosis Early-onset severe intrauterine growth Restriction) RCTs used daily SC doses of 75 mg SC for up to 10 weeks (~5,250 mg in total).^{15,16} In 2019, The Dutch STRIDER trial was stopped due to a non-significant increase in neonatal deaths and an increase in persistent pulmonary hypertension of the newborn in the SC group.¹⁷ Although a subsequent meta-analysis of data from 329 infants from the Australia-New Zealand, United Kingdom and Netherlands trials showed no difference in the risk of neonatal death or persistent pulmonary hypertension,¹⁸ the investigators recommend that SC is not prescribed outside of clinical trials.¹⁹ The indication and dosage of SC in the iSEARCH trial are however very different from the STRIDER trials as we will only use up to 150mg in women >37 weeks gestation.

1.1 Benefit/Risk Assessment

1.1.1 Risk Monitoring and Reporting

We will follow NHMRC recommendations (2018) for clinical trials involving therapeutic goods.²⁰ We will embed risk management into all aspects of the study protocol and trial design with early and ongoing risk assessment using risk indicators and thresholds and adjust monitoring activities based on issues and risks identified. Before starting, a risk monitoring plan will be agreed in the terms of

reference for the IDSMC and TMC. The NHMRC²⁰ identify two relevant risks pertinent to this RCT: those associated with the drug and with the conduct, design or methods of the trial.

Risk associated with the drug: This RCT fulfils criteria relevant to Type B risk category (Risk associated with modified use of an existing product). The trial manager will ensure traceability of the trial drugs to confirm full chain of custody from manufacture, bulk receipt, site distribution and storage and destruction after the study, to reconcile bulk supply against individual participant use. The IDSMC will be responsible for monitoring all Serious Adverse Events in the study protocol and will have the authority to recommend suspension of the study if major safety concerns arise.

Risks associated with trial conduct, design and methods: Consistent with Type B risk category we will perform central monitoring of key data, protocol adherence and data quality. Targeted site visits will be performed with frequency guided by central monitoring outputs. There will be regular central and remote monitoring of participant eligibility, rates of recruitment, withdrawals and losses to follow-up by site as well as compliance with all aspects of the protocol. Assessment of adverse event reporting rates between sites will be regularly undertaken. Assessment of case report forms (CRF) (e.g. review for consistency, late or poor CRF completion, checks to confirm CRFs have been completed by authorised personnel etc.) will be consistently done. The IDSMC will formally meet annually but will confer by tele/video conference if any concerns arise prior to the formal meetings.

1.1.2 Known potential risks

SC was initially used to treat hypertension and ischaemic heart disease and then widely used for erectile dysfunction. It is available on prescription in Australia and over the counter in other countries. It is also used in adult and newborn pulmonary hypertension.^{9 10} Our systematic review of 16 obstetric studies, including 165 patients from nine countries including UK, Iran, Egypt, India, Taiwan, Singapore, Ireland and Chile, found no long-term adverse maternal or perinatal outcomes.¹¹ Our recent Phase 2 RCT also did not demonstrate any differences in adverse outcomes between the placebo and SC cohorts.

1.1.3 Known potential benefits

If the study hypotheses are confirmed, wider implementation in other healthcare settings, particularly in low- and middle-income countries (LMIC) would also be likely, thus benefiting regions with the highest rates of perinatal morbidity and mortality. Intrapartum stillbirths in particular, account for 1 in 2 of all stillbirths, of which almost 98% occur in LMIC.²¹ Maternal deaths following CS are also disproportionately high – 1 in 4 deaths in LMICs occur in women who have undergone CS.²² If SC is shown to be of benefit, it will reduce the global health burden attributable to birth asphyxia which is a major cause of disability and comparable to congenital malformations, Type II diabetes or HIV/AIDS,²³ and may reduce maternal mortality rates.

Rationale for Sildenafil Citrate to reduce intrapartum hypoxia and fetal distress

Most cases of fetal distress in labour reflect some degree of pre-existing placental impairment²⁴ which reduces the capacity of the fetus to cope with the stress of uterine contractions.²⁵ In many term pregnancies, uterine blood flow can fall by 60% during contractions²⁶ provoking unexpected fetal distress. Fetal decompensation occurs²⁷ if there is insufficient time for placental reperfusion between contractions.²⁴ Fetal distress is a major factor in 1 in 4 emergency CS in Australia³ and is associated with significant perinatal morbidity and mortality (hypoxic ischaemic encephalopathy, cerebral palsy, intrapartum stillbirth and neonatal death).^{4 5 6} The only therapeutic options in response to fetal distress are emergency operative birth by CS or IVB – both of which have serious consequences including complications in future pregnancies.^{28,29} There is an urgent unmet need for effective interventions to reduce perinatal mortality due to intrapartum stillbirth³⁰ and improve maternal and neonatal outcomes by reducing fetal distress and emergency operative births in labour.

We³¹ have shown that a low fetal cerebro-placental ratio (CPR) on ultrasound identifies babies at increased risk of intrapartum fetal distress. We also showed that women with low pre-labour levels of placental growth factor (PIGF) are at similar risk.³² However, neither the CPR, PIGF nor computerised analysis of fetal heart rate patterns³³ or fetal ECG,³⁴ have been shown to improve important perinatal outcomes, further illustrating why this is a critical area of unmet need.

SC is a phosphodiesterase-5 inhibitor which enhances Nitric Oxide (NO) mediated effects by inhibiting cyclic guanosine monophosphate degradation. Increased bioavailability of NO improves vasodilatation and reduces vascular dysfunction secondary to vasoconstrictors and antiangiogenic factors.^{35,36} SC preferentially dilates pelvic blood vessels, increasing utero-placental blood flow.⁸ We postulate that SC improves uterine perfusion in labour, preventing fetal decompensation reducing adverse perinatal outcome due to fetal distress.

We completed a Phase 2 RCT¹² which tested if SC reduced the primary outcome of operative birth (CS or IVB) for fetal distress in term labour. SC reduced the relative risk of operative birth for fetal distress by 51% [(RR 0.49, 95% CI 0.33 - 0.73), p=0.0004; NNT = 5 (3-11)] and it also reduced pathological FHR patterns (15% vs. 32%; RR 0.48, 95% CI 0.31 - 0.75, p=0.0009) but it lacked power to show a significant effect on the composite neonatal outcome which was inconclusive with wide confidence intervals (20.7% vs. 21.3%; RR 0.97 (95% CI 0.62-1.50), p=0.89) and thus did not rule out major benefit or harm of this key endpoint.

In Summary

Given our promising Phase 2 results, iSEARCH will evaluate if intrapartum oral SC versus placebo reduces the relative risk of a composite of severe perinatal outcomes (Primary Outcome) related to intrapartum hypoxia by 35%, from 7% to 4.55%. Our overarching research questions are: Does maternal oral Sildenafil Citrate in labour reduce the primary composite endpoint of ten severe neonatal outcomes and secondary outcomes which include emergency operative birth for fetal distress, and is it more cost effective compared to placebo?

2 AIM AND OBJECTIVES

The aim of this study is to test the hypotheses that ≤ 3 doses of oral 50 mg SC vs. placebo will improve perinatal and/or maternal outcomes related to intrapartum hypoxia.

The primary objective is to demonstrate or refute a reduction of 35% in the relative risk of the composite perinatal outcome (from 7% to 4.55%) comprising ten components, shown below and in Table 1. Composite neonatal outcomes, appropriately constructed,^{37,38} are often employed in multicentre trials.^{39-43,44}

Composite primary outcome:

- I. Intrapartum stillbirth, or
- II. 28-day neonatal death, or
- III. Apgar score <4 at 5 minutes, or
- IV. Cord artery pH <7.0, or
- V. Neonatal encephalopathy, or
- VI. Neonatal seizures, or
- VII. Neonatal respiratory support lasting over 4 h, or
- VIII. Neonatal unit (Special Care or Intensive care) admission lasting over 48 h, or
- IX. Persistent pulmonary hypertension of the newborn, or
- X. Meconium aspiration

Table 1

Components of the primary outcome	Rate*	Association with long term adverse outcome
Intrapartum stillbirth ⁴⁵	0.1%	-
28 day neonatal mortality ⁴⁶	0.24%	-
Apgar score <4 at 5 minutes	0.5%	↑ risk of cerebral palsy ⁴⁷
Umbilical Cord artery pH <7.0	2.2%	↑ risk of HIE ⁴⁸
Neonatal encephalopathy	0.5%	↑ risk of death or disability ⁴⁹
Neonatal seizures	0.25%	↑ risk of death or disability ^{49,8}
Neonatal respiratory support for >4 h	3.6%	↑ risk of cerebral palsy ⁵⁰
Neonatal unit admission for >48 h	4.3%	↑ asthma after term respiratory morbidity ⁵¹
Persistent pulmonary hypertension	0.04%	↑ risk of death or disability ⁵²

Meconium aspiration	0.75%	↑ risk of death or disability ⁵³
Total (corrected for overlap)	7.0%	

**based on data from Grobman et al,⁵⁴ and the Australia New Zealand Neonatal Network⁵⁵*

Secondary objectives:

1. Reduces the relative risk of each individual component of the composite primary outcome (detailed in Table 1).
2. Reduces the relative risk of CS or IVB for fetal distress by 25% (from 20% to 15%).
3. Is cost-effective.

Tertiary objectives:

1. CS for suspected fetal distress
2. IVB for suspected fetal distress
3. CS for other indications
4. IVB for other indications
5. Spontaneous vaginal birth
6. Length of 1st and 2nd stage of labour
7. Post-partum haemorrhage (blood loss >1000ml)
8. Post-partum hysterectomy
9. Uterine rupture
10. Maternal Intensive Care Unit admission
11. Maternal mortality
12. Reduces the relative risks of cerebral palsy, cognitive delay and neurosensory disability at 2 years of age.

The correlative objectives of this study in a subset of participants are:

1. To determine if any fetal biophysical parameters (estimated fetal weight, umbilical artery pulsatility index, middle cerebral artery pulsatility index or cerebroplacental ratio) are predictive of efficacy of SC treatment.
2. To determine if maternal PIGF, sFlt-1 or sFlt-1:PIGF ratio are predictive of efficacy of SC treatment.

3 DESIGN

This study is a randomised, multicentre, two-arm parallel, phase III, double-blind, placebo-controlled trial.

4 STUDY POPULATION

Participants must meet all of the inclusion criteria and none of the exclusion criteria to be eligible for this trial. No exceptions will be made to these eligibility requirements at the time of registration. All enquiries about eligibility should be addressed by contacting the NHMRC CTC prior to registration.

4.1 Target Population

Pregnant women from 34⁺⁰ weeks planning a vaginal birth at term (≥ 37 weeks gestation). The research will be conducted in a manner sensitive to the cultural principles of Aboriginal and Torres Strait Islander communities.

4.2 Inclusion Criteria

1. Women with singleton or dichorionic twin pregnancies planning in-hospital vaginal birth at term (≥ 37 weeks gestation).
2. Age 18 years and older.
3. Willing and able to comply with all study requirements.
4. Signed, written informed consent. For Non-English speaking participants, a Health Care Interpreter must be engaged to obtain informed consent.

4.3 Exclusion Criteria

1. A woman should not be entered if the responsible clinician or the woman are, for any medical or non-medical reasons, reasonably certain that SC **would be inappropriate** for her in comparison with no treatment or some other treatment that could be offered in or outside the trial.⁵⁶
2. Triplets or higher order multiple births, which are generally delivered electively before term.
3. Contraindications to the investigational product SC (see 5.4).⁵⁷
4. Participants who are taking any type of nitrate drug therapy or who utilize short-acting nitrate-containing medications during labour [such as sodium nitroprusside, bosentan, fosamprenavir and ritonavir combination, hepatic enzyme inhibitors CYP3A4 (including itraconazole, ketoconazole, ritonavir, cimetidine, erythromycin, saquinavir, darunavir), or hepatic enzyme substrates (CYP3A4), medications used to treat pulmonary arterial hypertension, and other phosphodiesterase type 5 inhibitors like riociguat], due to the risk of potentially life-threatening hypotension.⁵⁷
5. Severe hepatic or renal impairment.⁵⁷

4.4 Study Enrolment

4.4.1 Screening

Written informed consent must be signed and dated by the participant, and signed and dated by the Investigator, prior to any study-specific screening investigations being performed.

All women will be screened for eligibility and approached by research midwives from 34⁺⁰ for consent to participate in the trial. Where appropriate, a researcher from the trial team will speak with the potential participant about the trial via a virtual/Telehealth platform, where e-signatures are to be integrated by participant and staff into the Participant Information Sheet and Consent Form. After informed consent, all women will be registered onto the online trial registration database. Once registered, all women will continue to receive routine obstetric care until the onset of spontaneous labour or induction of labour. Women will be advised to attend hospital when in labour according to local guidelines. Participation in this trial is not an indication for early presentation to hospital. Registration will be done according to the instructions in the Study Manual. Once the registration process has been completed, the participant will be assigned a study number. Individuals may only be registered once in this trial.

4.4.2 Registration and Randomisation

Women will be centrally randomised through the NHMRC CTC. Treatment allocation will be balanced using minimisation for the study site and women will be allocated to the treatment groups in a ratio of 1:1.

Registration, including randomisation, must be done according to the instructions in the Study Manual.

Randomisation will be performed after registration and consent to participate has been obtained when the participant is admitted to the hospital either in spontaneous labour or for induction of labour. It will be done according to the instructions in the Study Manual and should be undertaken after all screening assessments have been performed and the responsible investigator has verified the participant's eligibility. Women may only be randomised once in this trial. Once the randomisation process has been completed, the participant will be assigned a treatment arm.

4.5 Withdrawal of consent

Participants may withdraw consent at any time during the study without the need to justify their decision. Withdrawal of consent will not jeopardise future medical care.

Participants who do not receive study treatment for any reason will be encouraged to continue in the study. Please refer to section 6 regarding study procedures to be performed.

Participants who do not receive allocated treatment or withdraw from the study follow-up visits after randomisation will be included in the trial database and will be reported.

5 TREATMENT PLAN

5.1 Treatments

Sildenafil Citrate is the study intervention in this trial. Placebo is the control intervention. Treatment will commence after transfer to the labour ward once in labour or for induction of labour. The dose of SC is 50mg orally or equivalent placebo every 8 hours to a maximum of three doses.

5.1.1 Study Treatment

Participants will be randomised to one of the following two arms:				
Arm	Agent(s)	Dose	Route	Duration
Active	Sildenafil Citrate	50 mg	Oral	Every 8 hours from the start of labour or induction of labour to a maximum of 3 doses.
Control	Placebo	Placebo	Oral	

5.1.2 Required Background Treatment

All participants are expected to receive background treatment consistent with routine obstetric care.

5.2 Dose Modifications

No dose modifications are permitted.

5.3 Unblinding

Unblinding is not generally necessary for the management of a participant with an adverse event and, as it has an impact on the study's validity, is strongly discouraged. The need for unblinding should be very uncommon as the study intervention is rarely associated with severe side effects and it will not delay or prevent standard management of the participant. However, if required, it can be done after discussion with the Study Chair and NHMRC CTC Clinical Lead using the emergency unblinding number in the study manual.

5.4 Concomitant Medications

Any anti-hypertensive medication use will be recorded during admission for delivery

5.4.1 Prohibited

Participants who are on the following medications during labour (sodium nitroprusside, bosentan, fosamprenavir and ritonavir combination, hepatic enzyme inhibitors CYP3A4 (including itraconazole, ketoconazole, ritonavir, cimetidine, erythromycin, saquinavir, darunavir), hepatic enzyme substrates (CYP3A4), medications used to treat pulmonary arterial hypertension, and other phosphodiesterase type 5 inhibitors) are not eligible to participate in the study. Should treatment with any of these agents be required after recruitment, ongoing participation will need to be discussed with the Study Chair.

5.4.2 Concomitant Medication Reporting

Concomitant medications such as standard anti-hypertensives used in pregnancy will be recorded, however medications used to treat an SAE will not be recorded.

5.5 Compliance

Study Treatment compliance will be monitored and documented in the medical records and eCRFs.

5.6 Study Treatment Discontinuation

Women who satisfy the below criteria will, for their safety and wellbeing, have their treatment ceased:

- Acute severe maternal hypotension (<60/40 mmHg x3 episodes) sufficient to cause maternal and/or fetal compromise.
- Participant requests discontinuation from study.
- The investigative team believes it is in the best interest of the participant on grounds of safety or side effects or tolerability that discontinuation take place.

6 ASSESSMENT PLAN

6.1 Schedule of Assessments

Trial period	Maternal		Neonatal							Health Economic Evaluation
	Screening	Baseline	Onset of/during labour	At Birth or Discharge	Follow-up					
	≥ 34 ⁺⁰			28 days	6 months	12 months	18 months	24-36 months		
				Prior to Hospital Discharge	+/- 3 days	+ / - 1 month window if necessary				
Eligibility criteria ¹	X	X								
Informed consent ¹	X	X								
Registration ¹	X	X								
Randomisation ¹	X	X	X							
Clinical Assessment	X	X	X	X	X					
Continuous electronic fetal heart rate monitoring			X							
Cord blood gases				X						
Oxygen Saturation Screening				X						
Outcome Events		X	X	X	X				X	X
Adverse events of interest/SAE/SUSAR			X	X	X					
Concomitant medications reporting		X	X	X	X					
Maternal blood test for placental biomarkers ²		X	X							
Fetal ultrasound ²		X	X							
PBS MBS data										X
Record/confirm contacts for families				X	X	X	X	X	X	
Developmental assessments ³									X	

1. At any one timepoint
2. Only for participating Queensland sites
3. Subject to further funding

6.2 Details of Assessments

6.2.1 Blood Collection and Imaging

After registration but before the onset of labour, women at select participating Queensland hospitals, will have 20mL of blood collected into EDTA tubes for assay of sFlt-1 and PIGF levels.

In participating Queensland centres, an ultrasound will be performed by a member of the study team for:

- Estimated fetal weight
- Deepest liquor pocket
- Uterine Artery Pulsatility Index
- Umbilical vein flow
- Umbilical Artery Pulsatility Index
- Middle Cerebral Artery Pulsatility Index

Where possible, the ultrasound will be repeated 1-4 hours after administration of the first dose of the study medication to ascertain any changes in fetal and maternal circulation. The ultrasound data and maternal sFlt-1 and PIGF levels will allow post-hoc subgroup analysis to identify cohorts at risk of fetal distress that might derive greater benefit from SC treatment.

6.2.2 Management of Labour and Puerperium

All women routinely have their blood pressure taken when admitted to the labour ward. Blood pressure monitoring should follow local protocols and guidelines. Intrapartum FHR monitoring should follow local protocols and guidelines. Umbilical cord blood gases will be performed in all women. Classification of FHR abnormalities is based on the Royal Australian and New Zealand College of Obstetricians and Gynaecologists guidelines.⁴⁶ Management after birth will be in accordance with local hospital guidelines.

6.2.3 Management of the Infant after birth

All infants will also receive routine oxygen saturation screening as is current standard practice in Australia. This test complements the newborn examination and is used to detect hypoxemia in infants that otherwise appear well. It will be performed 24-48 hours after birth, although if necessary, it can also be done as early as 4 hours after birth. Infants with oxygen saturations of >95% are very unlikely to have major congenital heart or significant pulmonary disease, including persistent pulmonary hypertension of the newborn. Infants with oxygen saturations <95% will receive further assessment by the paediatric team and additional investigations including echocardiography may be required.

6.3 Follow-up After Treatment

Women and their newborns will be followed up for at least 28 days after end of treatment as described in Section 8.2 if adverse events are ongoing at the end-of-treatment visit.

Participants will be followed up by phone or text or email every 6 months until the study endpoint is reached. This is to ensure that their contact details remain up-to-date for the longer-term childhood neurodevelopment outcome study at 24-36 months (subject to further funding).

6.4 Follow-up after participant withdrawal

If a participant wishes to stop the 6-monthly contact, they will be asked for consent to allow information about their ongoing health status to be periodically collected via phone/email contact via a relative or their general practitioner (GP), or medical records, or the Australian Institute of Health and Welfare (AIHW) National Death Index (NDI).

For participants who have been lost to follow-up, Medicare may be used to provide updated contact information and information about any hospitalisations.

7 OUTCOMES, ENDPOINTS AND OTHER MEASUREMENTS

In-hospital perinatal and maternal outcome events occurring from screening to discharge home will be collected from medical records. With the exception of 28-day neonatal mortality, all outcome data will be routinely available before discharge. There are no formal study assessments visits. Research midwives will contact participants 28 days after discharge to ascertain if there any issues relating to the study that require additional follow up.

7.1 *Primary Composite Outcome*

A composite of the following adverse perinatal outcomes:

- Intrapartum stillbirth or
- 28-day neonatal mortality or
- Apgar score <4 at 5 minutes or
- Cord artery pH <7.0 or
- Neonatal encephalopathy (Sarnat score) up to 28 days or
- Neonatal seizures up to 28 days or
- Neonatal respiratory support >4h or
- Neonatal unit (Special Care or Intensive Care) admission lasting >48h or
- Persistent pulmonary hypertension of the newborn or
- Meconium aspiration

7.1.1 Intrapartum Stillbirth or 28-day Neonatal Death

Intrapartum stillbirth is the death in labour of an infant that was alive at the commencement of labour.

7.1.2 28-day neonatal mortality

Neonatal mortality is the death of an infant within 28 days of birth

7.1.3 Apgar score <4 at 5 minutes

Numerical score⁵⁸ to evaluate the infant's condition at 5 minutes after birth based on five characteristics – heart rate, respiratory condition, muscle tone, reflexes and colour. Assessed routinely and recorded in the medical record.

7.1.4 Cord artery pH <7.0

Significant metabolic acidosis defined as cord arterial blood pH <7.059

7.1.5 Neonatal encephalopathy

A clinically defined syndrome of disturbed neurologic function in the earliest post-natal days of life in an infant born at or beyond 35 weeks of gestation, manifested by a subnormal level of consciousness or seizures, and often accompanied by difficulty with initiating and maintaining respiration and depression of tone and reflexes.⁶⁰ Measured in infants >35 weeks using a modified Sarnat criteria⁶⁰ to assess severity of encephalopathy and graded from 0-3. Grade 0, no hypoxic ischaemic encephalopathy to grade 3, severe hypoxic ischaemic encephalopathy.

7.1.6 Neonatal seizures

Defined as the paroxysmal alteration in neurological function of behavioural, motor or neurological function. May be clinical or diagnosed on electroencephalogram.⁶⁰

7.1.7 Respiratory support lasting >4 hours

Supplementary respiratory support includes intubation and intermittent positive pressure ventilation, high flow oxygen via nasal cannula or CPAP.

7.1.8 Neonatal unit admission lasting >48 hours

Neonatal unit admission is defined as the number of hours from admission to discharge.

7.1.9 Persistent pulmonary hypertension of the newborn

Persistent pulmonary hypertension of the newborn is defined as a syndrome of failed circulatory adaptation at birth due to delay or impairment in the normal fall in pulmonary vascular resistance that occurs following birth. Infants with oxygen saturations of >95% are very unlikely to have persistent pulmonary hypertension of the newborn.

7.1.10 Meconium Aspiration

Presence of meconium in the trachea and symptoms/signs fulfilling criteria for persistent pulmonary hypertension of the newborn

7.2 Secondary Outcomes

7.2.1 Individual components of primary composite endpoints

These include individual components of the primary composite endpoint (Intrapartum stillbirth, 28-day neonatal mortality, Apgar score <4 at 5 minutes, Cord artery pH <7.0, Neonatal encephalopathy, Neonatal seizures, Neonatal respiratory support >4h, Neonatal unit (Special Care or Intensive Care) admission lasting >48h, Persistent pulmonary hypertension of the newborn or Meconium aspiration) of the primary composite outcome. There are ten items because outcomes (i) and (viii) each contain two individual components

7.2.2 CS or IVB for fetal distress

Defined as reduction in the relative risk of CS or IVB for fetal distress by 25% (from 20% to 15%).

7.2.3 Health Outcomes Relative to Costs

Cost effectiveness and budget impact of CS vs standard care in eligible populations. A cost-utility analysis will be undertaken from a payer's perspective with a one-year time horizon. A budget impact analysis will also be undertaken using a five-year time horizon. All costs will be presented in 2024/25 dollars, with no discounting applied. Further details are described in Section 8.2 below.

7.3 Tertiary Outcomes

7.3.1 CS for fetal distress

Defined as an emergency CS for fetal distress during labour as detected by fetal heart rate abnormalities based on the Royal Australian and New Zealand College of Obstetricians and Gynaecologists guidelines and/or abnormal fetal scalp pH or lactate levels.

7.3.2 Instrumental vaginal birth for fetal distress

Defined as a vaginal birth using forceps or vacuum extraction for fetal distress.

7.3.3 CS for other indications

Other indications include CS for failure to progress, failed induction of labour, malpresentation, failed instrumental birth, cord prolapse/presentation, uterine rupture etc.

7.3.4 Instrumental vaginal birth for other indications

Defined as a vaginal birth using forceps and vacuum extraction for non-fetal distress indications.

7.3.5 Spontaneous vaginal birth

Defined as the uncomplicated vaginal birth of an infant.

7.3.6 Length of 1st and 2nd stage of labour

The 1st stage of labour is defined as commencing when labour is diagnosed by the attending obstetric team to full dilatation (i.e. 10cm). The 2nd stage of labour is defined as the period of time from full cervical dilatation to birth of the infant.

7.3.7 Post-partum haemorrhage

Defined as excessive bleeding (>1000ml) following the birth of a baby.

7.3.8 Post-partum hysterectomy

Defined as the removal of the uterus (+/- cervix) after birth.

7.3.9 Uterine rupture

Defined as a full thickness tear through the uterine myometrium and serosa.

7.3.10 Maternal Intensive Care Unit admission

Intensive care unit admission is defined as the admission of the participant to an intensive care unit.

7.3.11 Maternal Mortality

Maternal mortality is defined as death of mother before discharge

8 SUBGROUP ANALYSES

Consistency of the treatment effect on the primary end point will be evaluated across six subgroups: (1) Small for gestational age (Birth weight <10th centile for gestation); (2) multiple pregnancy; (3) pre-eclampsia; (4) one or more previous CS; (5) non-cephalic presentation; (6) major fetal anomaly in this pregnancy.

8.1 Health Related Quality of Life

Upon registration to the study, consent will be sought for participation in the infant neurodevelopment follow-up portion of the study. If consent is given, contact details will be recorded and follow-up assessment will be performed as described below. If consent is not given or withdrawn before follow-up takes place (2-3 years corrected age), no further contact will take place.

Follow-up assessment for neurodevelopmental outcomes at 2-3 years corrected age will consist of the Ages and Stages Questionnaire (ASQ). To facilitate collection of 2-year study outcome data, 6 monthly contact by phone, email and/or post will be maintained with participants who have consented to regular contact and update of contact details. For women who have consented but subsequently lost to follow-up, Medicare information may be used to provide updated contact information. Research midwives will mail or email the ASQ to women that have consented to infant neurodevelopment follow up. Separate funding will be sought for this aspect of the study.

8.2 Economic Evaluation

A modelled cost-utility analysis will be undertaken using trial data. The analysis will be from a payer's perspective (covering government and individuals), and cover a one-year time horizon. A Markov microsimulation model will be constructed comparing treatment with SC with placebo (standard of care). Quality adjusted life years for will be assigned to each health state, and will be obtained from the literature⁶¹. Costs will include direct costs of the intervention (drug costs), as well as background health service use – hospital inpatient, emergency department, outpatient, community based care, and pharmaceutical use. All costs will be presented in 2023/24 dollars, with no discounting applied. We will seek consent to extract inpatient, outpatient, and emergency department data and Medicare Benefits Schedule and Pharmaceutical Benefits Scheme data for all participants from study entry to six months of age. Disutilities and costs associated with adverse events will be included. Parametric bootstrapping will quantify the uncertainty associated with the model parameters and 95% uncertainty ranges presented. A budget impact analysis will also be undertaken using a five-year time horizon.

9 SAFETY REPORTING

9.1 Definitions

An ADVERSE EVENT (AE) is any untoward medical occurrence in a patient or clinical investigational participant administered a pharmaceutical product and which does not necessarily have a causal relationship with this treatment. An AE can therefore be any unfavourable or unintended sign (including an abnormal laboratory finding), symptom, or disease temporally associated with the use

of a medicinal investigational product, whether or not considered related to the medicinal product (see below).

AEs include the following:

- All suspected adverse drug or device reactions
- All reactions from drug or device – overdose, abuse, withdrawal, sensitivity, toxicity or failure of expected pharmacological action (if appropriate)
- Apparently unrelated illnesses
- Worsening (severity, frequency) of pre-existing illnesses or symptoms
- Injury or accidents
- Abnormalities in physiological testing or physical examination that require clinical intervention or further investigation (beyond ordering a repeat examination)
- Laboratory abnormalities that require clinical intervention or further investigation (beyond ordering a laboratory test). Any untoward event that occurs after the protocol-specified reporting period which the Investigator believes may be related to the drug or device.

A SERIOUS ADVERSE EVENT (SAE) is any untoward medical occurrence that at any dose:

- results in death,
- is life-threatening (i.e. the participant is at risk of death at the time of the event),
- requires inpatient hospitalisation or prolongation of existing hospitalisation,
- results in persistent or significant disability or incapacity,
- results in a congenital anomaly/birth defect,
- other important medical events which, in the opinion of the investigator, are likely to become serious if untreated, or as defined in the protocol

NOTES:

- (i) The term “life-threatening” in the definition of “serious” refers to an event in which the participant was at risk of death at the time of the event; it does not refer to an event which hypothetically might have caused death if it were more severe.
- (ii) Important medical events which may not be immediately life-threatening or result in death or hospitalization but which may jeopardize the participant or may require intervention to prevent one of the listed outcomes in the definition above should also be considered serious.

In Australia the following definitions are used for reporting of safety events;

- A SIGNIFICANT SAFETY ISSUE (SSI) is defined as is a safety issue that could adversely affect the safety of participants or materially impact on the continued ethical acceptability or conduct of the trial. These events may be in addition to the current SAE/SUSAR reports and generally have a consequence related to participant safety within the current study protocol, which thus requires some type of amendment.
- An URGENT SAFETY MEASURE (USM) is one type of significant safety issue where sponsors or trial investigators act immediately to protect participants from an immediate hazard to their health and safety. USMs are often instigated before the TGA and HREC are notified. In these cases, it is strongly recommended that the sponsor contact the TGA within 24 hours of the measure being taken.

(Examples include:

- a serious adverse event that could be associated with the trial procedures and that requires modification of the conduct of the trial
- a hazard to the participant population, such as lack of efficacy of an IMP used for the treatment of a life-threatening disease
- a major safety finding from a newly completed animal study (such as carcinogenicity)
- a temporary halt/termination of a trial for safety reasons
- recommendations of the IDSMC, where relevant for the safety of participants, such as an increase in frequency or severity of an expected adverse reaction

- single case events (e.g. toxic epidermal necrolysis, agranulocytosis, hepatic failure) that leads to an urgent safety measure)

SSIs or USMs do not necessarily meet all criteria to be considered an SAE. For purpose of safety reporting, these events are to be reported as SAE with a note that this concerns an SSI or USM.

A **SUSPECTED UNEXPECTED SERIOUS ADVERSE REACTION (SUSAR)** is an SAE that is related to the drug or device and is unexpected, i.e. not listed in the investigator brochure or approved Product Information; or is not listed at the specificity or severity that has been observed; or is not consistent with the risk information described in the Participant Information Sheet and Informed Consent Form or elsewhere in the protocol. An event is causally related if there is a reasonable possibility that the drug [intervention] caused the AE, i.e. there is evidence to suggest a causal relationship between the drug and the event.

9.2 Recording Adverse Events

AEs will be recorded from the first dose of study treatment until 28-days post birth.

The following AEs of interest **will** be reported in the eCRF:

More frequent	Less frequent
Diarrhoea	Bronchitis
Dizziness	Cellulitis/Skin rash
Flushing	Epistaxis
Gastro-oesophageal reflux/dyspepsia	Hypo/paraesthesia
Headache	Migraine-like headache
Myalgia/Limb pain	Tremor
Vision changes (blurred, increased sensitivity)	
Vomiting	
Vaginal itch	

The following events **will be reported as SAEs**, regardless of whether they meet SAE criteria:

- Maternal Intensive Care Unit Admission
- Maternal hypotension (<60/40mmHg x3 episodes) requiring medical intervention
- Major postpartum haemorrhage >1,500 mls
- Maternal persistent visual changes requiring ophthalmic review
- Maternal death prior to discharge from hospital
- Neonatal admission to Neonatal Intensive Care Unit for >48 hrs
- Cord arterial pH <7.0
- Hypoxic ischaemic encephalopathy or neonatal seizure
- Stillbirth or neonatal death prior to discharge from hospital

9.3 Reporting of Serious Adverse Events (including SUSARs, SSIs and USMs)

The trial investigators are responsible for reporting all SAEs (including SUSARs), SSIs and USMs occurring during the study to the NHMRC CTC within 24 hours of investigational site staff becoming aware of the event according to the procedure documented in the Study Manual.

The NHMRC CTC will provide SUSAR, SSI and USM reports and SAE line listings to Investigators for submission to the lead Human Research Ethics Committee as required. The NHMRC CTC will submit 'reportable safety events' to the TGA in Australia in time to comply with the requisite specified regulatory time windows (usually 7 days for fatal/life threatening events with an 8 day follow-up report, and 15 days for other events).⁶²

10 CENTRAL REVIEW

10.1 Screening for eligibility

Research midwives/site investigators will follow pre-specified eligibility criteria detailed in the protocol. Refer to the Study Manual for screening procedures.

10.2 Imaging

Pre-treatment and post treatment ultrasound scans will only be performed in select Queensland hospitals. Refer to the Study Manual for scanning and collection procedures. .

10.3 Endpoint Adjudication

Any endpoint adjudication of outcomes will be reviewed by a central independent assessor blinded to both participant details and group assignment.

11 CENTRAL STORAGE OF BIOSPECIMENS

Blood samples will be collected from consenting participants at Queensland sites and shipped to Mater Pathology laboratory in South Brisbane for testing and storage. Samples will be sent using routine daily Mater Pathology courier services. Samples will be stored indefinitely or until used up unless otherwise arranged. Only laboratory researchers involved in the study will have access to samples and will not be able to link samples or data to personal information of patients. Participants retain the right to have their samples destroyed at any time by contacting their study doctor. If samples are destroyed, any data or analyses that were done before the request cannot be removed.

12 STUDY TREATMENT SUPPLY AND ACCOUNTABILITY

12.1 Study & Control Treatment

SC and placebo should be stored as recommended in the Study Manual. Full details on product handling information are provided in the Product Information and Pharmacy Manual. Both SC and Placebo will be prepared by a GMP-licensed manufacturer.

12.2 Drug Accountability

The Pharmacy Department at participating institutions will maintain a record of drugs dispensed for each participant and subsequent returns (if oral). The Pharmacy will also maintain a record of drug receipt and drug destruction as appropriate.

13 STATISTICAL CONSIDERATIONS

13.1 Sample Size

The incidence of the composite adverse outcome is estimated at 7% based on data in a recent large RCT in term pregnancies⁵⁴ and the Australian and New Zealand Neonatal Network. To detect a 35% reduction from 7% to 4.55% for an alpha of 0.05 and >80% power with a 10% drop-out in each arm, needs 3,200 women, also yielding >90% power to detect a 25% reduction for the secondary outcome (CS or IVB for fetal distress).

13.2 Statistical Analysis

Analyses of the primary and secondary outcomes will adhere to an intent-to-treat basis using generalised linear models (binary or normal). Intervention effect will be presented as relative risk (RR) or mean difference (MD), as appropriate, with 95% confidence intervals. Two-tailed P values <0.05 will be considered statistically significant. Primary analyses will be unadjusted. Numbers needed to treat to prevent one adverse outcome will be calculated. Where there are differences in the baseline characteristics between the two treatment groups that might be associated with outcomes, secondary analyses of the primary outcome will be carried out using multiple (log-binominal) regression. Reporting will follow CONSORT and TRIPOD guidelines.

The key secondary outcomes for iSEARCH comprise the individual components of the primary composite outcome (Section 2.2.1 above). P values adjusted for the ten comparisons performed for this set of endpoints will be derived using the Benjamin-Hochberg procedure, with a family-wise error rate of 5%.^{63,64} Results of other endpoint, subgroup and sensitivity analyses will be interpreted in

proper context and with due consideration of the risk of Type I error. The clinical evidence of the statistical evidence from iSEARCH will take into account the recommendations of Pocock et al.^{65,66}

13.3 Interim analyses

The Independent Data and Safety Monitoring Committee (IDSMC) will review interim data on the primary outcome, specified adverse events and other evidence after 50% of recruitment or whenever they deem appropriate informed by recommendations of Peto, Pocock and others.^{67,69} There will be no adjustment to alpha for interim analyses using a Haybittle-Peto boundary.

Interim analyses of the primary composite outcome: The IDSMC will advise the TMC if in their view there is proof beyond reasonable doubt of net benefit or harm for the primary composite outcome, employing a commonly used formal threshold of $P < 0.001$ for nominal significance, as recommended by Geller and Pocock.⁶⁹

Interim analyses of mortality: The IDSMC will advise the TMC if in their view there is a difference in mortality due to intrapartum stillbirth and/or 28-day neonatal mortality identified as a deviation from the null indicated by a Haybittle-Peto boundary which is equivalent to $P < 0.0027$,^{67,68} which would be needed to justify recommending early stopping.

14 STUDY ORGANISATION and COMMITTEES

14.1 Study coordination

The study is a locally developed and led investigator initiated collaborative group study. Coordination, monitoring, data acquisition and management and statistical analysis will be performed by the NHMRC CTC.

14.2 Trial Management Committee

The Trial Management Committee (TMC) will oversee study planning, monitoring, progress, review of information from related research, and implementation of recommendations from other study committees and external bodies (e.g. ethics committees).

The TMC will consider recommendations from the IDSMC about whether to continue the study as planned, modify, or stop it, based on interim analyses or other information. The TMC will consist of all Chief Investigators and chaired by Prof Sailesh Kumar.

14.3 Independent Safety and Data Monitoring Committee

The IDSMC will be established by the TMC. Agreed terms of reference⁷⁰ for the IDSMC will be developed with the TMC. Guided by its Charter, the IDSMC will provide independent assessment of participant safety and trial progress, making recommendations to the TMC about the continuation of the trial based on data made available by the trial statistician. The IDSMC will comprise an interdisciplinary group with relevant expertise.

The IDSMC will be supplied with:

- Cumulative list of SAEs on a 2-monthly basis (or sooner if requested)
- Interim analysis report(s)
- Relevant information about the study's progress including accrual rates, compliance, etc.
- Medical, ethical or other information that might influence decisions to continue the study

15 ADMINISTRATIVE ASPECTS

15.1 Ethics and regulatory compliance

In Australia, the study will be conducted according to the ICH Guideline for Good Clinical Practice Integrated Addendum to ICH E6 (R1): Guideline for Good Clinical Practice ICH E6(R2) dated 9 November 2016 annotated with TGA (Therapeutic Goods Administration) comments and in compliance with applicable laws and regulations. The study will be performed in accordance with the National Statement on Ethical Conduct in Human Research 2007 (updated 2018 and as amended

from time to time), the Australian Code for the Responsible Conduct of Research (2018, and as amended from time to time), and the principles laid down by the World Medical Association in the Declaration of Helsinki 2013.

To this end, no patient will be recruited to the study until all the necessary approvals have been obtained and the patient has provided written informed consent. Further, the investigator shall comply with the protocol, except when a protocol deviation is required to eliminate immediate hazard to a participant. In this circumstance the NHMRC CTC, principal investigator and HREC must be advised immediately.

15.2 Confidentiality

The study will be conducted in accordance with applicable Privacy Acts and Regulations. All data generated in this study will remain confidential. All information will be stored securely at the NHMRC CTC, University of Sydney and will only be available to people directly involved with the study.

15.3 Protocol deviations and amendments

A protocol deviation is any noncompliance with the clinical trial protocol, GCP, or other ethical or regulatory requirement. The noncompliance may be either on the part of the participant, the investigator, or the study staff. As a result of deviations, corrective actions are to be developed by the sponsor and implemented promptly. All deviations must be addressed in study source documents, reported to the CTC. Protocol deviations should be sent to the Lead HREC and RGO as per their local guidelines.

Changes and amendments to the protocol can only be made by the TMC. Approval of amendments by the Institutional HREC is required prior to their implementation. In some instances, an amendment may require a change to a consent form. The Investigator must receive approval/advice of the revised consent form prior to implementation of the change. In addition, changes to the data collected, if required, will be incorporated in the amendment.

The investigator should not implement any changes to, or deviations from, the protocol except where necessary to eliminate immediate hazard(s) to trial participant(s).

In response to unforeseen circumstances (e.g. natural disasters, pandemics), the study investigators may direct an immediate change to the protocol with the primary aim of continuing health care and ensuring the participant's best interests at all times. Changes will be communicated with the relevant parties e.g. the Lead HREC/RGOs, and regulatory authorities if appropriate, and participants in due course.

15.4 Data Handling and Record Keeping

All trial data required for the monitoring and analysis of the study will be recorded on the eCRFs provided. All required data entry fields must be completed. The eCRF serves as part of the investigator's record of participants' study-related data. Data corrections will be done according to the instructions provided. The investigator will be asked to confirm the accuracy of completed CRFs by signing key CRFs as indicated.

Source documents pertaining to the trial must be maintained by investigational sites. Source documents may include a participant's medical records, hospital charts, clinic charts, the investigator's participant study files, as well as the results of diagnostic tests such as X-rays, laboratory tests, and electrocardiograms. The investigator's copy of the case report forms serves as part of the investigator's record of a participant's study-related data.

The following information should be entered into the participant's medical records:

- a. Participant's name, contact information and protocol identification.
- b. The date that the participant entered the study, and participant number.
- c. A statement that informed consent was obtained (including the date).
- d. Relevant medical history
- e. Occurrence and status of any adverse events.
- f. The date the participant exited the study, and a notation as to whether the participant completed the study or reason for discontinuation.

Data will be stored on a dedicated trial database, on an encrypted, password protected server at the NHMRC Clinical Trial Centre. All study-related documentation will be maintained for 15 years following completion of the study. Only the trial coordinator, trial statistician and Principal Investigator will have access to the data.

15.5 Study Monitoring

Data from this study will be monitored by Clinical Trials Program staff from the NHMRC CTC or their delegates in accordance with a separate study monitoring plan. This will include centralised review of CRFs and other study documents for protocol compliance, data accuracy and completeness. Monitoring may include monitoring visits to investigational sites during the study for source data verification, review of the investigator's site file and drug handling records. The NHMRC CTC will be given direct access to source documents, CRFs and other study-related documents. By signing the informed consent form, the participant gives authorised NHMRC CTC staff direct access to their medical records and the study data.

15.6 Audit and Inspection

This study may be participant to audit or inspection by representatives of the CTC or representatives of local e.g. the TGA.

15.7 Clinical Study Report

A Clinical Study Report which summarises and interprets all the pertinent study data collected will be issued which may form the basis of a manuscript intended for publication. The Clinical Study Report or summary thereof will be provided to the Human research Ethics Committees (HRECs) that approved the study and the funding agency (MRFF).

15.8 Publication and Data Sharing Policy

The Trial Management Committee will appoint a Writing Committee to draft manuscript(s) based on the trial data. Manuscript(s) will be submitted to peer-reviewed journal(s). The Writing Committee will develop a publication plan, including authorship, target journals and expected dates of publication. All publications must receive written approval from the TMC prior to submission.

The CTC will ensure that a summary of the clinical trial results is submitted to an appropriate register no later than 1 year after the completion date (referred to as the "primary completion date") of the clinical trial, which is defined as the date of final data collection for the primary outcome measure.

15.9 Conflict of Interest and Financial Disclosure

Any actual conflict of interest of persons who have a role in the design, conduct, analysis, publication, or any aspect of this trial will be disclosed and managed. Furthermore, persons who have a perceived conflict of interest will be required to have such conflicts managed in a way that is appropriate to their participation in the trial. The study leadership in conjunction with the CTC and collaborative group has established policies and procedures for all TMC members and site investigators to disclose all conflicts of interest and will establish a mechanism for the management of all reported dualities of interest. Investigators are responsible for providing information on financial interests during the course of the study.

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